

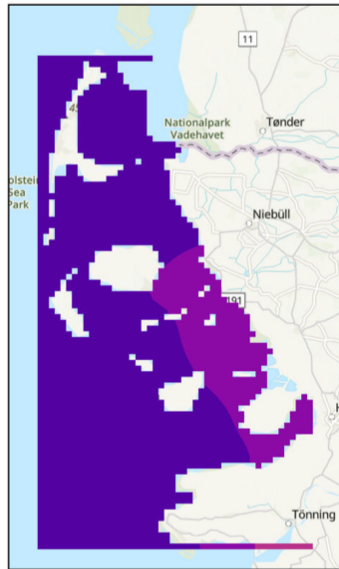


Map 3.1. Seagrass classification based on remote sensing of sentinel satellite of 17/07/2018

Chosen surface model NO3 concentration in the Wadden Sea

Surface model of NO3 concentration using spline method (tension)

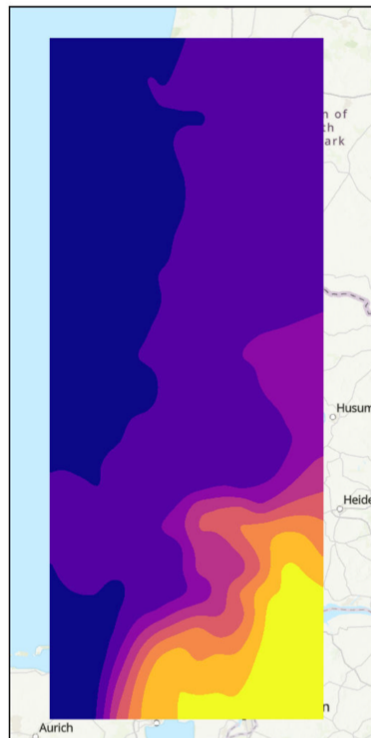
Surface model of NO3 concentration using spline method (tension): study area only



0 5 10 20 Kilometers



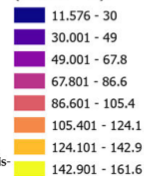
GeoBasis-DE/LVermGeo SH, Esri, TomTom, Garmin, Foursquare, FAO, METI/NASA, USGS, Esri, CGIAR, Geodatasbyrølsen, USGS, Esri, Geodatasbyrølsen, NASA, NGA, USGS, GeoBasis-DE/LVermGeo SH, Esri, TomTom, Garmin, FAO, NOAA, USGS, Esri, USGS



0 15 30 60 Km

Legend

NO3 concentration (mol m⁻³)



Map created on 17/10/24
Made by: Group 2
Coordinate system:
WGS-1984-UTM-Zone32N
Data sources:
Nutrients_NO3,
NO3_sample_points,
NO3_validation_points

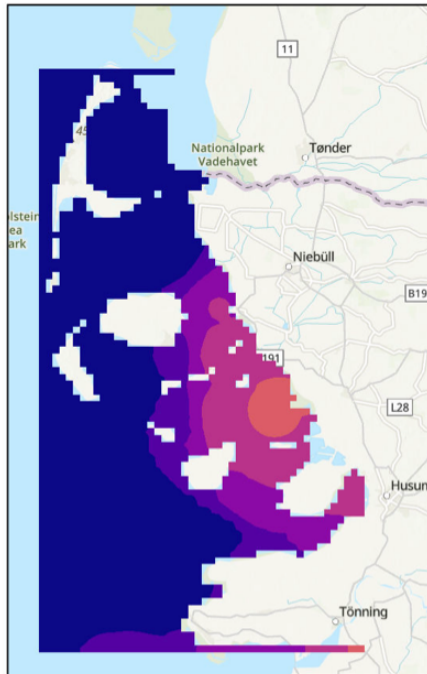
Map 3.2 Chosen surface model (spline) NO3 concentration in the Wadden Sea

PO4 concentration

The estimated mole concentration of PO4 in sea water for three different surface models using three different interpolation methods (IDW 12 points, IDW 4 points, and spline) were plotted against the validation values of mole concentration of PO4 for the given validation points (**Figure 3.2**). The root mean square error (RSME) was calculated for each of the three interpolation methods. The RSME for IDW with 12 points was 0.0456, the RSME for IDW with

Chosen surface model PO4 concentration in the Wadden Sea

Surface model of PO4 concentration using IDW method with 4 points: in study area only

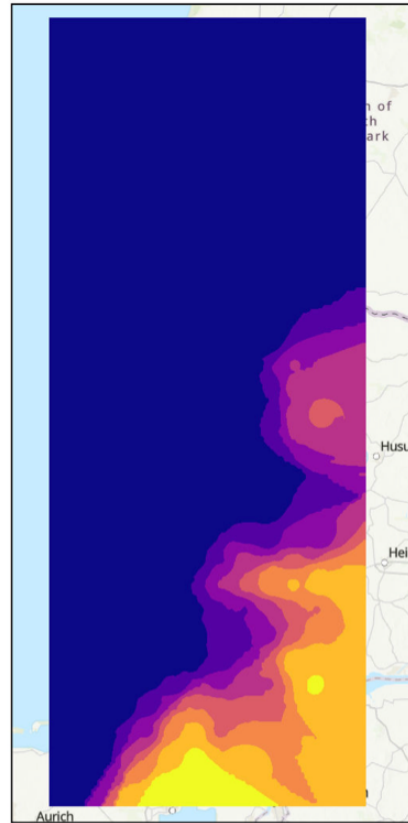


0 5 10 20 Kilometers



Geobasis-DE/LiVermGeo SH, Esri, TomTom, Garmin, Foursquare, FAO, METI/NASA, USGS, Esri, CCJAR, Geodatasystemen, USGS, Esri, Geodatasystemen, NASA, NOAA, USGS, Geobasis-DE/LiVermGeo SH, Esri, TomTom, Garmin, FAO, NOAA, USGS, Esri, USGS

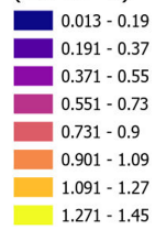
Surface model of PO4 concentration using IDW method with 4 points



0 15 30 60 Km

Legend

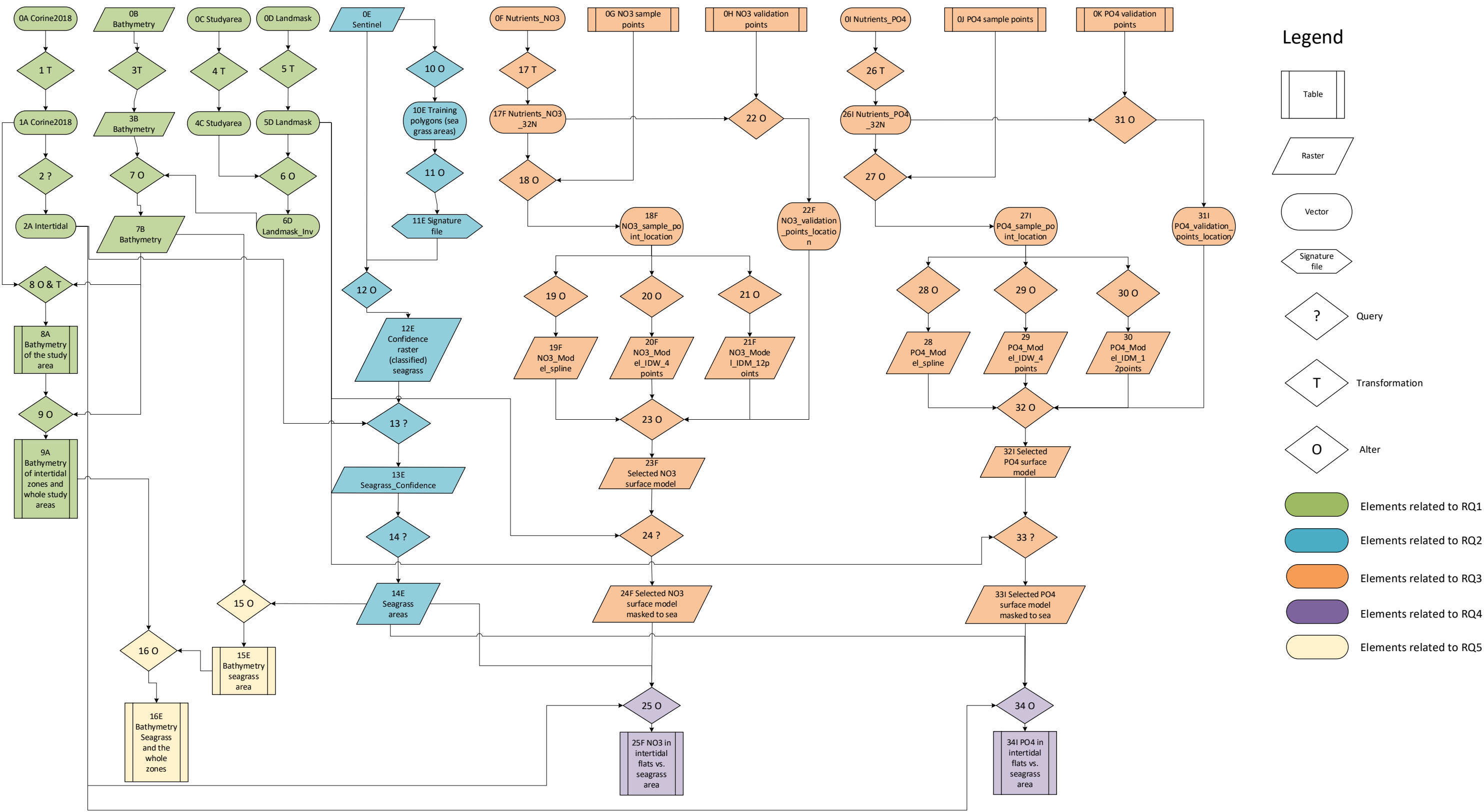
PO4 concentration
(mol m⁻³)



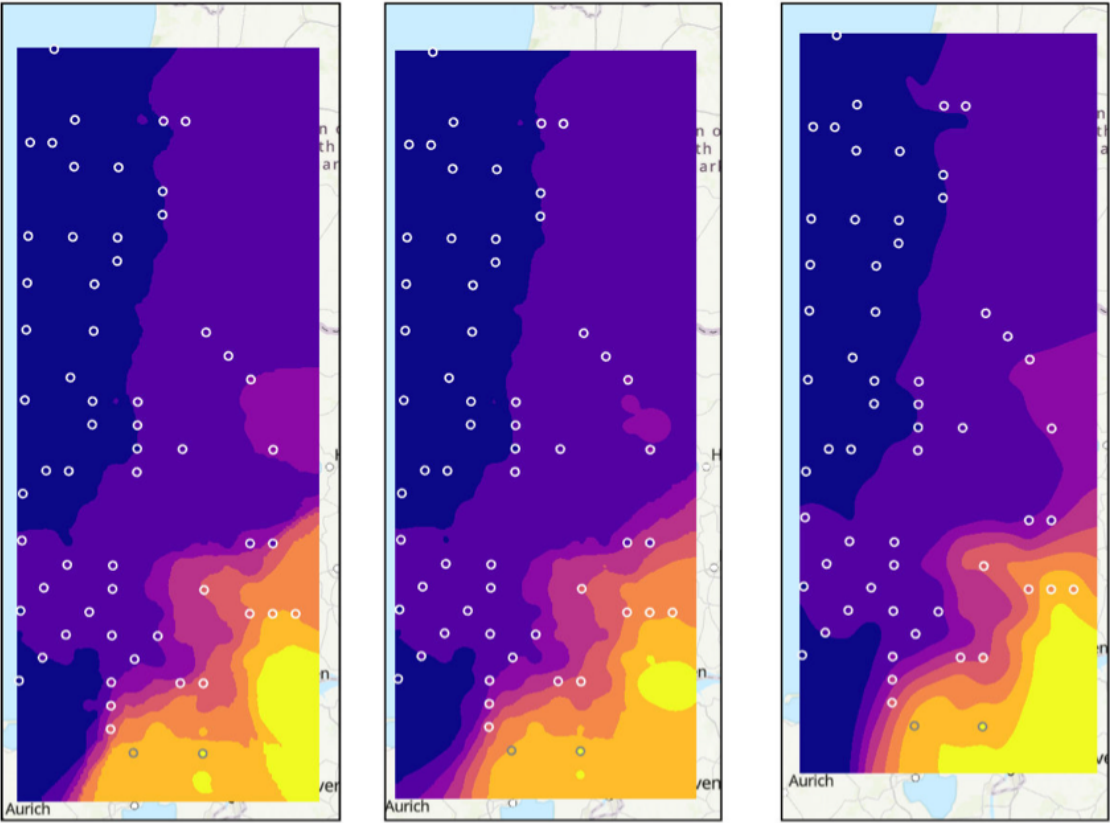
Map created on 17/10/24
Made by: Group 2
Coordinate system:
WGS-1984-UTM-Zone32N
Data sources:
Nutrients_PO4,
PO4_sample_points,
PO4_validation_points

Map 3.3 Chosen surface model (IDW 4points) PO4 concentration in the Wadden Sea

Annex 1. Data action model



Comparison of three interpolation methods of NO3 concentration with validation point values of NO3 concentration in the Wadden Sea



IDW with 4 points

Spline

IDW with 12 points

Legend

Concentration of NO3
(mol m⁻³)

- 11 - 30
- 30.001 - 49
- 49.001 - 67.8
- 67.801 - 86.6
- 86.601 - 105.4
- 105.401 - 124.1
- 124.101 - 142.9
- 142.901 - 161.6

Concentration of NO3
(mol m⁻³)

- 11.576 - 30
- 30.001 - 49
- 49.001 - 67.8
- 67.801 - 86.6
- 86.601 - 105.4
- 105.401 - 124.1
- 124.101 - 142.9
- 142.901 - 161.6

Map created on 17/10/24
Made by: Group 2
Coordinate system:
WGS-1984-UTM-Zone32N
Data sources:
Nutrients_NO3,
NO3_sample_points,
NO3_validation_points

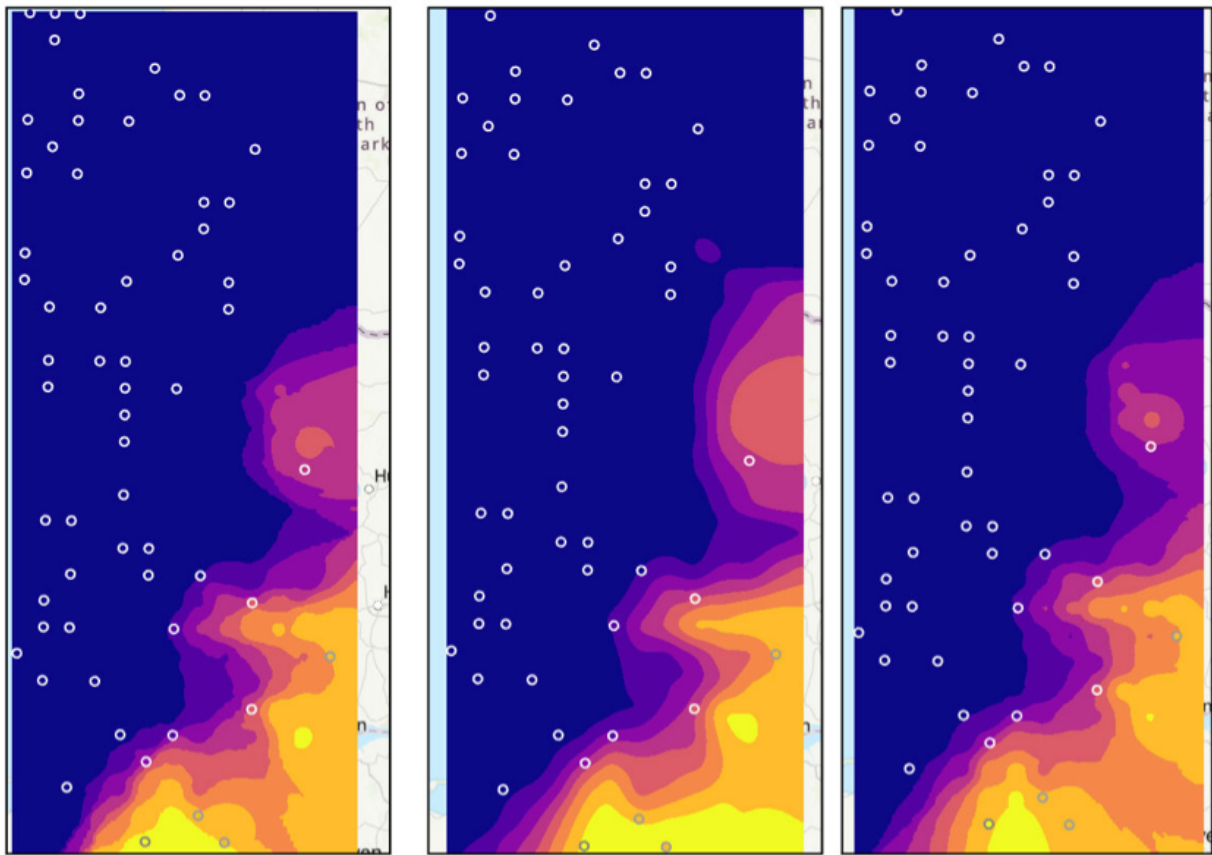


Esri, CGIAR, USGS, GeoBasis-DE/LVermGeo SH,
Esri, TomTom, Garmin, FAO, NOAA, USGS



Validation points are coloured with the same colour scale. In this way, differences between surface model estimated values and true validation values can be seen visually

Comparison of three interpolation methods of PO4 concentration with validation point values of PO4 concentration in the Wadden Sea



IDW with 4 points

Spline

IDW with 12 points

Legend

0 25 50 100 Kilometers

PO4 concentration
(mol m⁻³)

- 0.012 - 0.19
- 0.20 - 0.37
- 0.38 - 0.55
- 0.56 - 0.73
- 0.74 - 0.90
- 0.901 - 1.09
- 1.091 - 1.27
- 1.271 - 1.45

PO4 concentration
(mol m⁻³)

- 0.013 - 0.19
- 0.191 - 0.37
- 0.371 - 0.55
- 0.551 - 0.73
- 0.731 - 0.9
- 0.901 - 1.09
- 1.091 - 1.27
- 1.271 - 1.45

Map created on 17/10/24
Made by: Group 2
Coordinate system:
WGS-1984-UTM-Zone32N
Data sources:
Nutrients_PO4,
PO4_sample_points,
PO4_validation_points



Esri, CGIAR, Geodatastyrelsen, USGS, Esri, CGIAR, USGS,
GeoBasis-DE/LVermGeo SH, Esri, TomTom, Garmin, FAO,

Validation points are coloured with the same colour scale. In this way, differences between surface model estimated values and true validation values can be seen visually